9 (a) The nth term of a sequence is $3n + 4$

Explain why 21 is not a term of this sequence.

5th term: 3 × 5 + 4 = 19; 6th term: 22.

21 lies between these consecutive terms.

2 marks total
Both neighbouring terms justify why 21 is missing

(2)

(b) Here are the first three terms of a different sequence.

1 2 4

Write down two numbers that could be the 4th term and the 5th term of this sequence.
Give the rule you have used to get your numbers.

4th term = 8; 5th term = 16.

Rule: double the previous term.

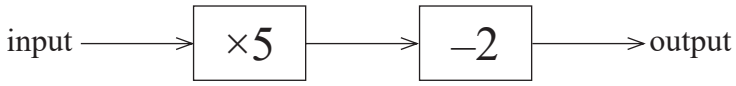
1 mark
Two consistent next terms

1 mark
Rule matches all five terms

(2)

(Total for Question 9 is 4 marks)

10 Here is a number machine.



(a) Work out the **output** when the input is 8

(1)

(b) Work out the **input** when the output is 28

(2)

(Total for Question 10 is 3 marks)

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